Stagflation and Asset Management

PANASHE BERA, CAMPBELL R. HARVEY, KRISTINA HOOPER*

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ABSTRACT

We analyze asset performance across inflation and growth states using 107 years of monthly US data. We construct a systematic three-regime taxonomy classifying each month as Overheating, Stagflation, or Contained (our baseline) using real-time inflation and growth signals, unlike prior work relying on fixed inflation thresholds or ex-post NBER recession dating. Both high-inflation states produce poor returns: the 60/40 portfolio earns -1.3% p.a. in real terms in Overheating and -0.2% in Stagflation, versus 9.0% in Contained periods. We also condition on whether inflation is accelerating or decelerating. Under our six-month specification, accelerating inflation in Overheating is associated with materially weaker equity and bond returns than decelerating inflation, though this is sensitive to the measurement horizon. In Stagflation, these differences are generally less pronounced. This contrast may partly reflect timing: by the time Stagflation is confirmed, some deterioration may already be priced in and markets may be anticipating recovery.

Keywords: Stagflation, overheating, inflation hedging, macroeconomic regimes, regime classification, 60/40 portfolio.

JEL codes: E31, E32, G11

*Harvey is at Duke University, the National Bureau of Economic Research and is an adviser to Man Group plc. The other authors are at Man Group plc. The authors would like to thank Alex Preston and participants at the Man Alternative Investment Symposium for valuable comments. Please direct correspondence to panashe.bera@man.com.

1. Introduction

Inflation and asset returns have a complicated relationship. For example, in the depths of a recession, a positive inflation shock may be viewed as good news; it is a sign that the economy is recovering. However, in a period of normal growth, the same inflation surprise could have a very negative effect on asset prices. Our paper analyzes the impact on asset returns of the interaction between inflation and economic growth. We focus on two challenging high-inflation environments for asset managers: Stagflation and Overheating.

We argue that the standard approach of looking at asset sensitivities to inflation surprises can be misleading if growth regimes are not taken into account. Further, for asset managers it is crucial to know, in real time, what growth regime you are in. For example, some look at sensitivities in recessions versus non-recessions relying on the dating system of the National Bureau of Economic Research (NBER). However, NBER recessions are not dated in real time. The average reporting lag to date the end of the last four recessions was 18 months.[1] Our second contribution is to propose a real-time method of determining growth and non-growth periods based on a long-term series of industrial production. Our inflation data is also sufficiently lagged so that the declaration of Stagflation or Overheating states occurs in real time.

Our third contribution focuses on another understudied dimension of inflation – acceleration. Inflation is the rate of change in the CPI level. Asset returns are influenced by changes in *expected* inflation. That is, if an inflation shock is considered a one-time event, it will not necessarily change expectations or asset prices. Acceleration is likely correlated with changes in expectations and we examine its impact on asset prices.

The timing of these changes matters. By the time high inflation and weak growth are visible in the economic data, markets may already have priced in substantial deterioration and begun to anticipate disinflation or recovery. Returns during a confirmed Stagflation regime can therefore differ materially from the losses incurred as the economy moves towards it.

We seek a regime definition that is transparent, a history long enough to contain materially different inflation episodes, and a broad panel of assets to provide a richer study of cross-sectional performance dynamics. Our goal is to provide asset managers with new analysis on the cross-section of inflation-adjusted asset returns in different high-inflation environments.

Our work builds on a body of research that links macro regimes to asset returns. Neville et al. (2021) study the best strategies for inflationary times using roughly a century of data, and find that commodities, trend-following strategies and certain long-short equity style factors can protect real wealth while nominal bonds and equities often struggle during periods of elevated inflation. Their inflation surge regime is defined by a fixed 5% inflation threshold combined with inflation acceleration. We consider a similar pool of assets as in their study, but we differ in a way

[1] Troughs: April 2020 (announced July 19, 2021), June 2009 (announced September 20, 2010), November 2001 (announced July 17, 2003), and March 1991 (announced December 21, 1992). See NBER.

that matters for portfolio construction: we consider the state of the economic growth alongside inflation dynamics. Inflation with strong growth is not the same problem as inflation with a contracting economy, and the diversifying assets that help differ across the two states.

The growth dimension is central to the relatively small Stagflation literature. Baltussen et al. (2023) extend the sample back to 1875 across several countries and show weak real equity and bond outcomes in Stagflation, while several factor premia remain positive. Where we differ is in the growth measure: they identify Stagflation using recessionary periods which are defined retrospectively by the NBER, whereas we use industrial production relative to its own trailing trend in a real-time framework, while acknowledging that historical macroeconomic data are subject to revision. Cervi and Iqbal (2024) offer the closest methodological precedent by grading post-war Stagflation episodes using time-varying inflation and growth benchmarks. We share the trailing-window philosophy, apply it to monthly industrial production from 1919, and evaluate a broader cross-section of asset returns. Our analysis of three historically noteworthy "stagflationary" episodes identifies themes that are consistent with the results from Cieslak and Pflueger (2023), who viewed demand-shock-induced inflation as "good" and cost-push (supply-driven) inflation as "bad".

We offer several insights. First, we construct a three-regime classification that accounts for a conservative two-month macroeconomic data delay and confirms a state change only after three consecutive observations to limit noisy regime transitions (our regime confirmation filter). Second, we evaluate the performance of 47 asset and strategy returns and seek to identify any material differences in performance between assets and across regimes. Third, we condition the final regimes on whether inflation is accelerating or decelerating and examine the dispersion in performance across individual episodes.

High inflation is associated with weaker average real returns for traditional assets relative to Contained periods. Conditioning on whether inflation is accelerating or decelerating sharpens the Overheating comparison further, where accelerating inflation coincides with weaker equity and bond outcomes. Stagflation is more heterogeneous: gold, commodities, trend and equity factors sometimes provide positive returns, but their performance varies substantially across independent episodes.

The remainder of the paper proceeds as follows. **Section 2** describes the data and methodology. **Section 3** sets out the classification framework. **Section 4** reports regime classification and asset performance results. **Section 5** considers the direction of inflation, while **Section 6** presents the independent Stagflation episode evidence and historical context. **Section 7** offers some conclusions, discusses potential implications for portfolios and sets out the principal limitations of our study. **Appendix A** reports the results for the same study conducted for the UK and Japan and **Appendix B** provides the US data summary and sensitivity checks.

2. Data and Methodology

The raw US dataset spans approximately 127 years of monthly observations, from January 1899 to May 2026. After the seasonal-adjustment and rolling-threshold warm-ups described below, the return analysis covers January 1919 to May 2026. We construct a seasonally adjusted US Consumer Price Index (CPI) series from a long-dated, non-seasonally adjusted series sourced from Finaeon. To avoid introducing lookahead bias into our classification framework and thereby developing a long-dated inflation signal that could have been constructed in real time, we derive seasonal adjustment factors using a trailing 10-year lookback window. For completeness, we calculated the correlation between month-on-month percentage changes in our own seasonally adjusted US CPI series with the official seasonally adjusted series published by the Bureau of Labor Statistics (BLS) over a common window (February 1947 to May 2026); the correlation between the two CPI series was 0.92. Further detail on the seasonal-adjustment construction is provided in the appendix.

The regime classification uses rolling 10-year reference windows. We therefore retain 10 years of warm-up data from 1899 to 1909 to construct the long-dated seasonally adjusted CPI series, and a further 10 years from 1909 to 1919 for the rolling macroeconomic thresholds. The return analysis consequently begins in January 1919.

The macroeconomic series are a seasonally adjusted US Consumer Price Index and the US Industrial Production Index (our growth proxy), both sourced from Finaeon. Industrial production is an imperfect stand-in for total output, since it understates services, but it has a consistent monthly history back to the early 1900s and it correlates strongly with GDP over comparable horizons. US GDP, by contrast, has no reliable monthly analogue and a relatively short quarterly history from 1947.

While we advocate real-time measures of inflation and growth, revisions make the measures quasi-real time. The literature shows that the correlation between revised and original release data is around 0.95 for four-quarter changes in real GDP and 0.98 for 12-month growth in industrial production (Aruoba, 2008). Revisions to CPI are irrelevant because revisions only impact the seasonal weights. Given we are measuring 12-month inflation, the correlation is 1.0 by construction.

The asset universe comprises 47 return series spanning core asset classes, Kenneth French 12-industry portfolios, commodities, real estate, trend-following strategies and long-short equity factors. The core analysis focuses on the US, with UK and Japanese evidence reported in the appendix. Returns are expressed in local currency terms and deflated by the relevant seasonally adjusted price index (CPI for the US and Japan and RPI for the UK).

We consider a number of different long-short US equity factors in our study: the “Small Minus Big” (SMB) factor; the “High Minus Low” (HML) factor; the “Robust Minus Weak” (RMW) factor; the “Conservative Minus Aggressive” (CMA) factor; the Momentum (WML) factor; the “Quality

Minus Junk" (QMJ) factor; and the "Betting-Against-Beta" (BAB) factor. The SMB and HML factors are as explained in Fama and French (1993), RMW and CMA factors as per Fama and French (2015) and the WML factor follows the approach outlined in Carhart (1997). The QMJ and BAB factors are as derived by Asness et al. (2019) and Frazzini and Pedersen (2014) respectively.

The paper also examines two real estate indices. The first is US REITs index, which is represented by the FTSE Nareit All Equity REITs Total Return Index, and the second is the case-Shiller US House Price Index. The latter is a price index, so does not capture the impact of income generated by underlying real estate assets. We have used monthly data for both indices.

Three construction choices merit clarification. Commodities are represented by futures returns only, with no spot-price backfill; each series begins when futures data becomes usable. Early spot histories for gold contain long runs of administered or pegged prices and are not informative about tradable returns. Gold is therefore represented by COMEX futures from February 1975, after the Bretton Woods peg had ended, and silver begins in 1963. Gold has 616 monthly observations, including 90 Stagflation, 68 Overheating and 458 Contained months. **Appendix B** reports effective start dates and regime coverage for every return series.

Fixed income returns are constant-maturity approximations, rather than published investable-index returns. For the 2-, 5-, 10- and 30-year nominal Treasuries, and the Aaa-rated and Baa-rated corporate bonds, monthly return combines prior-month yield carry, a duration term, and a second-order convexity term. Note, for the two corporate bond series, we have not incorporated any assumptions relating to historical default and recovery rates, so the series represent synthetic carry plus duration-and-convexity effect implied return series with no loss adjustments in respect of defaults. The 3-month Treasury bill monthly return is its annualized yield divided by 12. This captures the principal carry and yield-repricing channels, but not index rebalancing, transaction costs or other implementation details of a live index.

The 10-year TIPS series applies the same carry plus duration-and-convexity impact approximation. Our 10-year real yield series is Goldman Sachs' backcast[2] from March 1959 through December 1996 spliced with Bloomberg's generic 10-year TIPS yield from January 1997.

Trend-following uses the all-asset trend series of Neville et al. (2021), which followed the methodology of Hamill et al. (2016) who constructed time-series momentum strategies across asset classes using liquid futures and forward data. The strategies in those papers had volatility targets of 15% p.a. We rescale this to a 10% annualised volatility target to make its risk scale more comparable with that of a diversified institutional allocation.

[2] Marshall (2020) describes Goldman Sachs' construction: 112 economic and financial inputs are reduced by partial least squares to ten factors. Observed 10-year TIPS yields since 1997 are then regressed on those ten factors and the fitted relationship is projected backwards.

3. Regime Classification Framework

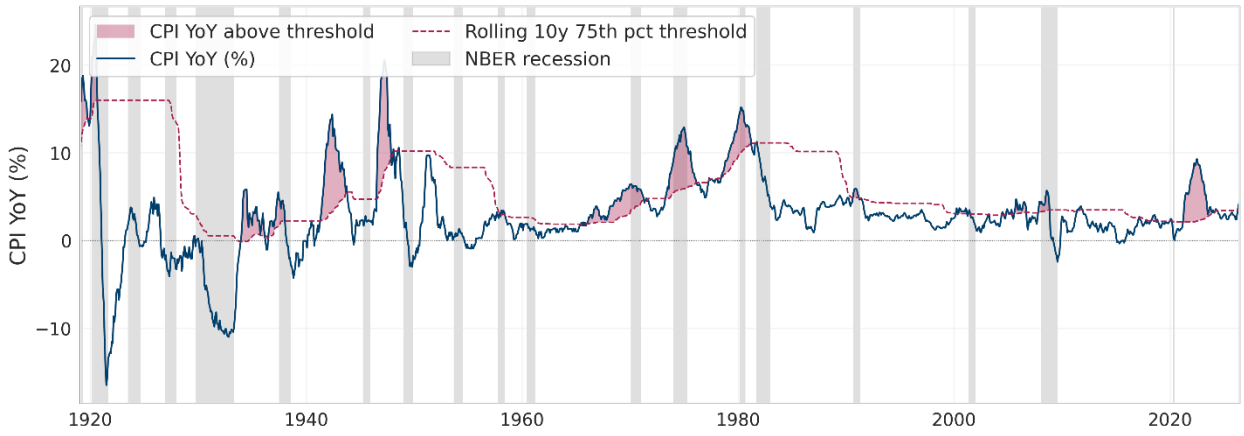
3.1 Signal construction

We build the taxonomy from two binary signals defined on year-over-year percentage changes. Both the "high" inflation and "weak" growth signals are calculated relative to a trailing 10-year history rather than a fixed threshold chosen over the full sample. To account for the publication delay associated with CPI and industrial production data, we apply a conservative two-month lag to both series.

Months are labelled as "high-inflation" when year-on-year CPI inflation exceeds the 75th percentile of its own trailing 10-year distribution. The use of a rolling window is particularly relevant for the inflation signal, given structural changes in how central banks manage price pressures (see **Exhibit 1** below). An increased focus on keeping inflation low and stable by the Fed in the 1980s (the onset of the "Great Moderation") meant that inflation volatility after the 1980s was significantly lower than during the preceding 60-year period; as such, 5% inflation in the 1940s was very different from 5% inflation in 2022.

The rolling construction uses only information dated at or before each observation and is consistent with a real-time implementation. It can, however, produce a lower threshold in low-inflation eras. We address that concern through robustness checks using different inflation percentile thresholds and trailing reference-window lengths.

Exhibit 1. Year over year CPI rate vs. rolling 10-year 75th percentile



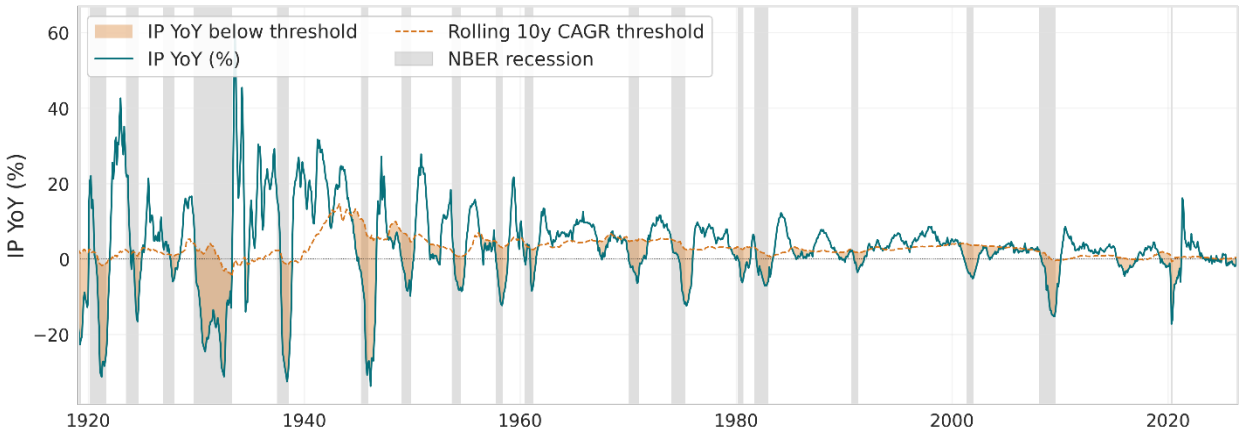
Source: Man Group, Finaeon. Red shading = YoY inflation is above 75th percentile. Grey bars = NBER recessions. Jan 1919 to May 2026.

Months are labelled as "weak-growth" when the year-on-year industrial production growth rate falls below its trailing ten-year compound annual growth rate (CAGR). We could have opted to implement a stricter weak-growth signal, e.g., by only considering periods where growth was negative or below the 25th percentile (for symmetry), however we settled on the trailing 10-year

CAGR threshold because it allows us to identify not just low economic growth environments, but also periods where growth may still have been positive but was weakening; indeed, in many cases we would likely see informative movements in asset prices as growth weakens, before we necessarily see growth turn negative. So, the two signals are deliberately asymmetric.

We also analyzed alternatives to the 10-year CAGR rule. Under the same two-month delay and three-month confirmation rule, our rule identifies 172 Stagflation months with average year-on-year IP growth of -1.6% p.a. A trailing mean identifies 183 months with IP growth of -1.2%, whereas a trailing median identifies 215 months and produces positive average year-on-year IP growth (+0.3%). A stricter trailing-25th-percentile rule and a fixed zero-growth rule identify 102 and 94 months respectively and concentrate on much deeper contractions. The CAGR appears to be a useful middle ground: it identifies economically weak activity relative to trend without reducing the episode sample to only outright contractions.

Exhibit 2. Annual Industrial Production growth vs. rolling 10-year CAGR



Source: Man Group, Finaeon. Orange shading = YoY growth is below rolling CAGR threshold. Grey bars = NBER recession. Jan 1919 to May 2026.

3.2 Three-regime taxonomy

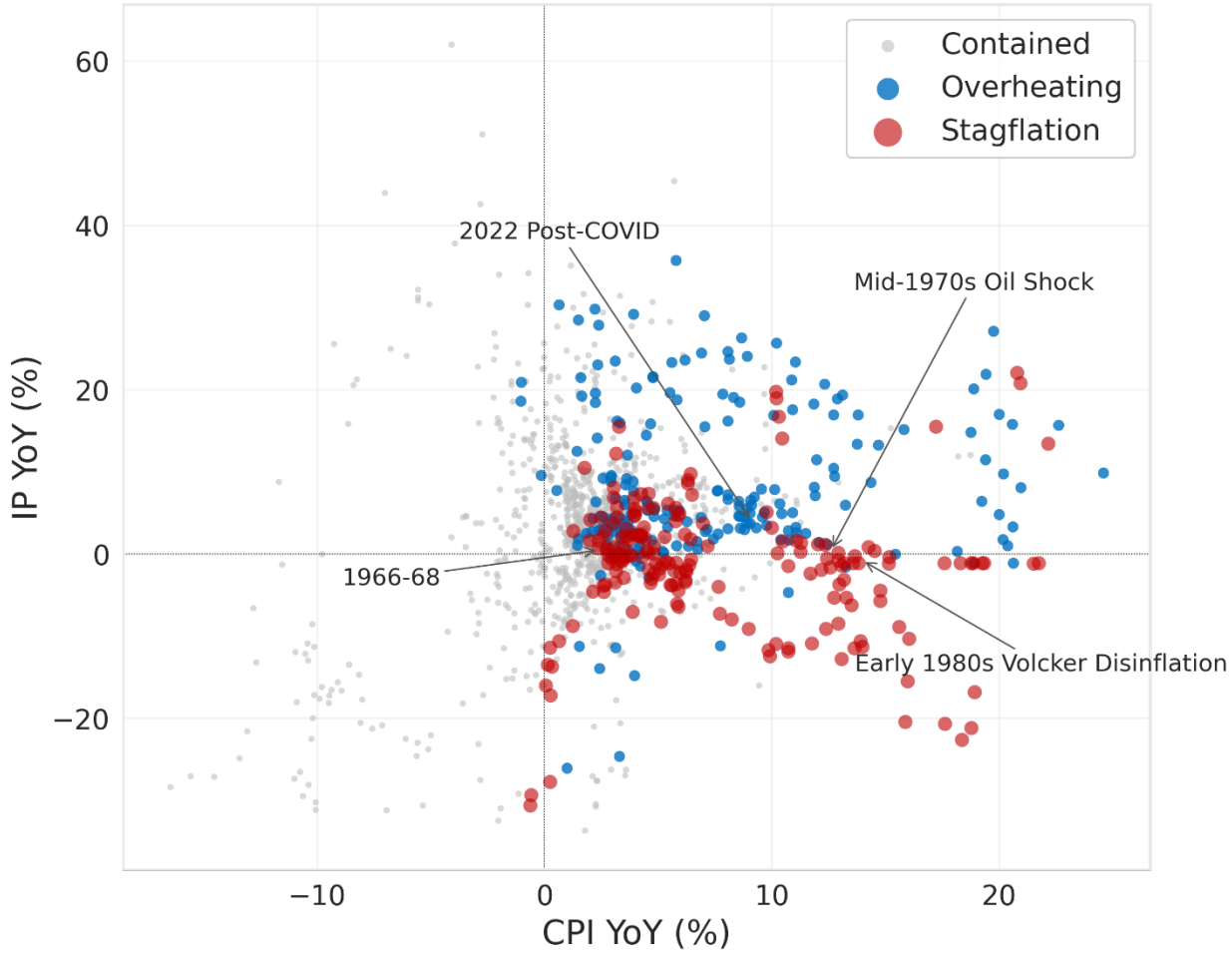
Together, the two signals generate the regimes directly. High inflation with weak growth is Stagflation while high inflation with healthy growth is Overheating. Every month without high inflation is assigned to the Contained baseline, regardless of its growth state.[3]

To limit noisy changes in the raw classification, we apply a three-month confirmation filter. A raw state must appear in three consecutive months before it is finalized. The first two candidate months retain the previously confirmed state, and the new state begins only in the third month; no earlier month is relabelled after confirmation. Together with the two-month macro-data delay,

[3] It is possible to segment Contained into two regimes: Non-elevated Inflation+Weak Growth and Non-elevated Inflation+Healthy Growth. However, our research focuses on high-inflation states.

this ensures that the classification framework avoids lookahead bias to the best of our abilities, acknowledging that the macroeconomic data is subject to revisions.

Exhibit 3. Growth x Inflation state space



Source: Man Group, Finaeon. Jan 1919 to May 2026.

Exhibit 3 shows that Overheating months (high inflation with above-trend growth) cluster in the upper-right quadrant, while Stagflation months cluster in the lower-right quadrant. The mid-1970s oil shock and the early 1980s Volcker disinflation period lie deep in the Stagflation quadrant, whereas post-pandemic inflation in 2022 lies in the Overheating quadrant. Borderline observations illustrate why the confirmation filter is needed.

4. Results

4.1 Regime Classification Results

The Contained baseline accounts for 72.2% of the sample. Overheating and Stagflation account for 14.4% and 13.3%, respectively.

Exhibit 4. Regime distribution and macro characteristics

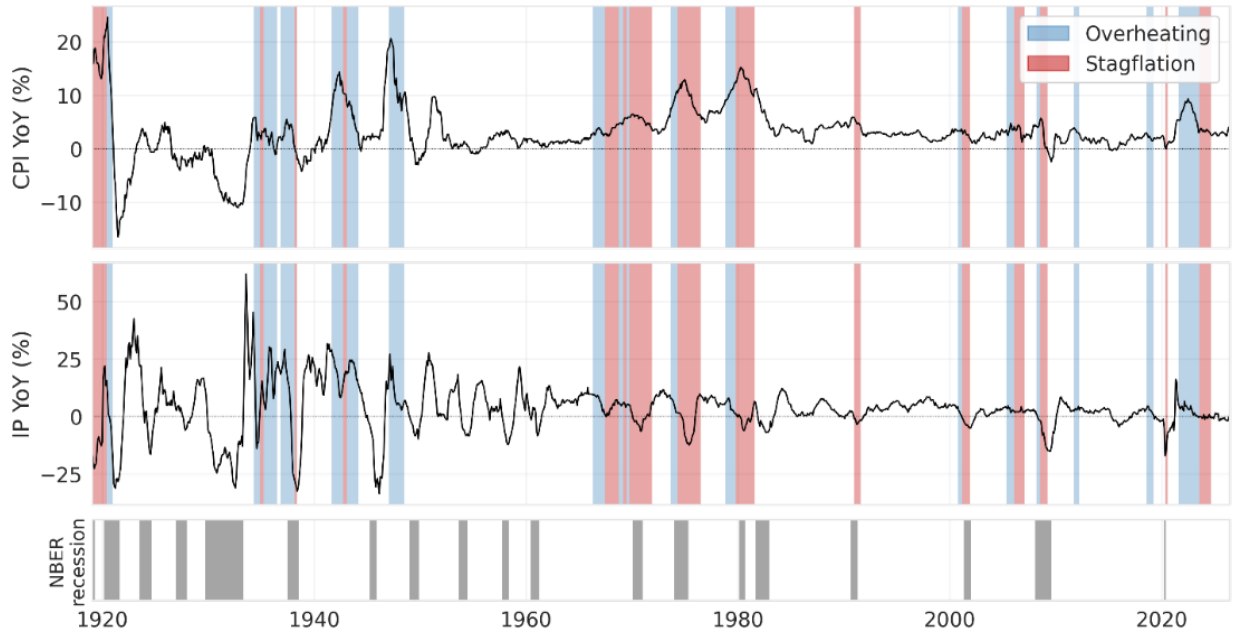
Regime	Months	Share	Avg. YoY CPI	Avg. YoY IP	Recession overlap*
Overheating	186	14.4%	6.7%	8.0%	12%
Stagflation	172	13.3%	7.2%	-1.6%	34%
Contained	931	72.2%	1.5%	3.2%	17%
Total	1,289	100.0%	3.0%	3.3%	18%

Source: Man Group. 3-month regime confirmation filter applied to regime classification framework. *Recession overlap measures the percentage of months in each regime that are classified as NBER recession months. Jan 1919 to May 2026.

As is evident in **Exhibit 4**, the two inflation regimes are well-separated on growth, which is the point of the design. Both carry high inflation, around 7% p.a. on average, but Overheating pairs it with strong industrial production growth of 8.0% p.a., while Stagflation pairs it with a contracting industrial economy at -1.6% p.a. The recession overlap confirms the distinction: 34% of Stagflation months fall inside NBER recessions, versus 12% for Overheating months.

The 172 Stagflation months comprise 15 separate episodes, while the 186 Overheating months comprise 18. The longest Stagflation episodes are 1969-71 (26 months), 1974-76 (26 months) and 1979-81 (21 months). The post-pandemic inflation surge of 2021-22 is classified as Overheating because industrial production was still strong during the economic reopening; the 2023-24 period is classified as Stagflation once growth weakened and inflation remained elevated.

Exhibit 5. Annual CPI and IP with Overheating and Stagflation regimes shaded



Source: Man Group, Finaeon. Blue = Overheating, Red = Stagflation. Grey bars in bottom panel show NBER recession periods. Jan 1919 to May 2026.

4.2 Asset Returns by Regime

We next compare average real returns across the three regimes. Equities and bonds are the core reference assets because of their prominence in institutional portfolios; the broader panel adds commodity futures, equity factors and trend-following strategies to examine the scope for diversification.

We first examine average real returns to visualise how well these asset classes were able to withstand the inflationary pressure during different regimes.

Exhibit 6. Summary of inflation-adjusted asset performance across different regimes

Panel A. Equities, Bonds and the 60/40 Portfolio

Asset	Start Date	Average Real Returns (% p.a.)		
		Stagflation	Overheating	Contained
Equities				
S&P 500	Jan-1919	0.7%	-0.2%	12.4%
S&P 500 (from Jul 1926)	*Jul-1926*	*1.2%*	*0.2%*	*12.0%*
Consumer Non-Durables	Jul-1926	5.3%	-0.5%	10.7%
Consumer Durables	Jul-1926	1.8%	0.5%	15.2%
Manufacturing	Jul-1926	0.9%	1.9%	13.0%
Energy Equities	Jul-1926	5.7%	9.6%	10.1%
Chemicals	Jul-1926	3.1%	-1.0%	12.1%
Business Equipment	Jul-1926	1.6%	2.2%	15.0%
Telecoms	Jul-1926	-2.9%	-2.5%	10.6%
Utilities	Jul-1926	-0.7%	0.8%	10.6%
Wholesale and Retail	Jul-1926	9.5%	-0.4%	11.6%
Healthcare	Jul-1926	5.7%	-2.1%	13.0%
Finance	Jul-1926	3.6%	1.8%	11.7%
Other Industries	Jul-1926	-0.9%	-0.8%	10.4%
Bonds				
US T-bills	Jan-1919	-0.6%	-2.0%	1.4%
US 2-Year Treasury	Jan-1941	1.7%	-4.1%	1.2%
US 5-Year Treasury	Jan-1941	0.9%	-4.6%	2.0%
US 10-Year Treasury	Jan-1919	-1.6%	-2.9%	3.9%
US 30-Year Treasury	Jun-1953	-1.4%	-6.4%	5.3%
US 10-Year TIPS	Apr-1959	1.8%	2.4%	2.8%
US Aaa Corporate	Jan-1919	-1.8%	-1.5%	4.7%
US Baa Corporate	Feb-1919	-1.2%	-1.5%	6.4%
Diversified				
60/40 (S&P 500/ 10-Year Treasury)	Jan-1919	-0.2%	-1.3%	9.0%

Panel B. Alternative assets

Asset	Start Date	Average Real Returns (% p.a.)		
		Stagflation	Overheating	Contained
Commodities				
Broad Commodities (EW)	Apr-1929	-2.0%	5.2%	9.1%
Gold	Feb-1975	9.4%	14.2%	7.7%
Silver	Jul-1963	5.3%	25.0%	11.2%
Precious Metals	Jul-1963	-3.8%	21.6%	12.3%
Agriculture	Aug-1929	-0.2%	3.5%	7.0%
Energy Futures	Apr-1983	-23.8%	14.0%	15.7%
Softs	Apr-1929	2.7%	2.9%	9.2%
Livestock	Jan-1965	5.7%	-0.5%	9.7%
Industrial	Aug-1959	-8.4%	8.7%	11.8%
Real Estate				
US REITs	Jan-1972	-0.4%	6.3%	10.2%
US House Price Index	Apr-1953	-0.8%	0.0%	1.4%
Trend-Following				
Multi-Asset Trend	Mar-1928	13.5%	7.4%	14.0%
FX Trend	Apr-1971	2.2%	2.2%	10.9%
Equity Trend	Mar-1928	5.2%	2.6%	5.6%
Commodity Trend	Feb-1947	14.1%	7.0%	11.9%
Bond Trend	Apr-1962	7.9%	10.0%	8.2%
Long-Short Equity Factors				
Multi-Factor Equities (EW)	Jul-1963	7.1%	2.5%	5.0%
Equity Momentum (WML)	Jul-1963	8.7%	-0.6%	9.3%
Equity Value (HML)	Jul-1963	3.8%	5.1%	3.9%
Equity Profitability (RMW)	Jul-1963	6.3%	4.5%	2.5%
Equity Investment (CMA)	Jul-1963	5.0%	3.4%	3.0%
Equity Quality (QMJ)	Jul-1963	6.6%	4.5%	3.8%
Equity Size (SMB)	Jul-1963	8.0%	-0.4%	1.8%
Equity Bet-Against-Beta (BAB)	Jul-1963	11.2%	0.7%	10.5%

Source: Man Group, Finaeon, Bloomberg, Kenneth R. French Data Library, Shiller Data Library, Goldman Sachs, AQR. Jan 1919 to May 2026.

Exhibit 6 provides a heatmap of the cross-section of average inflation-adjusted returns. Traditional assets generally have weaker average real returns in the two inflationary regimes than in Contained periods, while the range of outcomes across assets is wider.

It is important to emphasize that the results above are based on arithmetic averages of monthly real returns within each growth-inflation state. As such, they potentially mask significant heterogeneity in performance outcomes across independent Overheating and Stagflation episodes. Indeed, when we consider the 15 individual Stagflation episodes identified by our real-

time signal, the distribution of *annualized compounded* real returns for equities and 10-year Treasuries is quite broad (especially relative to the Overheating state); the 10th and 90th percentiles for equities were -28.4% and 36.8%, and -13.1% and 10.7% for 10-year treasuries respectively. Point estimates are therefore helpful, but not sufficient for drawing definitive conclusions about cross-sectional performance dynamics across different regimes.

This heterogeneity is exhibited more clearly in **Section 6** where we examine the performance of different asset classes across three historically noteworthy Stagflation episodes.

Equities

Exhibit 6 shows that inflation, whether Stagflation or Overheating, is generally bad for equities. The S&P 500 earns an inflation adjusted 12.4% p.a. in Contained periods, compared with -0.2% in Overheating and 0.7% in Stagflation. Theoretically, inflation impacts both the discount rate as well as the future cashflows. The increase in the discount rate depresses stock prices. If future cashflows are lower due to low growth, that further reduces prices. One might think that an overheating economy might come with an increase in expected cashflows, thereby offsetting the discount rate effect. In our sample, average equity returns are slightly lower in Overheating than in Stagflation. This comparison depends partly on when deteriorating conditions enter market prices relative to our regime classification. A confirmed Stagflation period may include both further losses and a recovery as investors anticipate improving conditions, even while reported growth remains weak. Our regime averages combine these phases and do not identify how much bad news was already reflected in prices.

Additional insight is found by examining sector returns. The Contained regime average returns are higher than the high-inflation regimes across every sector. That said, in Stagflation regimes there are some industries that exhibited comparatively minimal underperformance, such as Wholesale and Retail and, to a lesser extent, Healthcare. It is also the case that there can be significant differences between Stagflation and Overheating by sector. For example, Healthcare delivers a 5.7% return in Stagflation and a -2.1% return in Overheating.

It is worth reflecting on what **Exhibit 6** does and does not show. This exhibit would look somewhat different if we used ex-post dating of stagflationary episodes. However, such knowledge is not available to investors. Indeed, some of our real-time Stagflation months would not be classified as Stagflation under an ex-post-dating approach. We argue this type of false positive is not a weakness of our approach but a strength. We are providing an analysis of asset returns conditional on the real-time information.

Bonds

Nominal bonds have the most direct inflation exposure due to the fixed nature of their contractual cashflows. These cashflows are discounted by a rate that reflects real interest rates, expected inflation and an inflation risk premium. A positive inflation shock can raise expected inflation, thereby increasing the discount rate and lowering bond prices. Longer maturities are exposed to the full term structure of inflation expectations. The worst scenario is when the entire term

structure shifts upward. Shocks that are perceived to be short-lived have a minimal impact on long-term bond prices.

Similar to equities, real returns during Contained regimes are higher for all nominal bonds (the 2-year treasury is the only exception). Performance is poorest in the Overheating regime for all nominal bonds except the synthetic Aaa-rated corporate bond series. Here discount rates may be increasing because of a surge in inflation expectations coupled with an increase in the expected real rate, which covaries positively with economic activity.

Corporate bond prices also respond to changes in credit spreads (the difference between the corporate bond yield and duration-matched nominal Treasuries), as markets reprice the credit worthiness of corporate borrowers. As credit spreads widen, corporate bond yields increase relative to nominal Treasuries, which has an adverse impact on their price. Credit spreads, especially those of lower-rated corporate bonds, often widen during periods of elevated market and economic uncertainty, which might partially explain their poor performance during Stagflation and Overheating periods. Indeed, based on our analysis sample, the average monthly change in spreads[4] for Baa-rated corporate bonds was +1.0bps in the Overheating state and +2.4bps in Stagflation, compared to -0.7bps in the Contained state. The larger average spread widening exhibited in the Stagflation state is consistent with markets pricing in a weaker growth environment and therefore a higher risk of corporate default. It is important to note that the synthetic corporate bond series do not capture historical corporate default and recovery experience, so they potentially overstate historical performance experience; given that both series reflect higher quality, investment grade corporate bonds, we expect this impact to be minimal.

TIPS provide protection in both high-inflation regimes. The returns are not significantly different from the Contained regime. Their 2.4% and 1.8% real returns in Overheating and Stagflation respectively are less impaired than those of nominal bonds, but somewhat lower than their 2.8% Contained-period average. This is consistent with protection rather than immunity.

Not surprisingly, the 60-40 portfolio struggles in both high-inflation environments: the real returns are -1.3% in Overheating and -0.2% in Stagflation, against 9.0% in Contained periods. Equities and nominal bonds diversify many shocks, but high and changing inflation can damage both sleeves together.

Alternative assets

A commodity can be part of the shock that raises the price level, but the return on a futures contract also reflects inventories, supply response, global demand, storage and carry. Strong activity can support demand-sensitive commodities in Overheating, while weak activity can reverse that support in Stagflation. This distinction helps explain why the broad equal-weighted commodity series earns 5.2% in Overheating and -2.0% in Stagflation, even though both are high-

4 For simplicity, we have estimated spreads approximately as the difference between Aaa- and Baa-rated corporate bond yields and 10-year nominal Treasury yields. This is an illustrative, approximate measure which does not take into account differences in duration between the corporate bond and Treasury series and the effect of options embedded in the corporate bonds.

inflation states. It also explains why a broad allocation can obscure very different bucket exposures.

Gold and silver have a different economic role than other commodities and often resemble currencies. They can respond to real-rate declines, currency concerns and demand for stores of value, but these drivers need not coincide in every inflation episode. Gold earns 14.2% in Overheating and 9.4% in Stagflation, while the precious metals bucket earns 21.6% and -3.8%. The difference is due to both variation in the available history and the composition of the precious metals bucket; precious metals include platinum and palladium, commodities often classified as precious metals despite also possessing strong industrial (and therefore demand-sensitive) characteristics.

The results for real estate assets are mixed. US REITs produced strong average real returns of 6.3% p.a. during Overheating periods, consistent with the notion that real estate can potentially act as a hedging asset during inflationary periods. Performance during Stagflation periods, however, is relatively poor, with US REITs delivering an average real return of -0.4% p.a., similar to other long-duration equity sectors such as Telecoms and Utilities. The US House Price Index, our proxy for unlisted US real estate, returned -0.8%, 0.0% and 1.4% p.a. real on average during Stagflation, Overheating and Contained periods respectively. It is worth noting that this index is a price-based index, so does not capture income received from the underlying real estate assets.

Trend can respond to persistent price moves across asset classes, including falling bonds or rising commodity markets, rather than requiring a forecast of inflation. The broad Trend-Following series is positive in both high-inflation states; 13.5% in Stagflation and 7.4% in Overheating. In fact, the average trend return in Stagflation is only 0.5% below the performance in the Contained state. Looking at the underlying trend sleeves, equity, bond and commodity trend provide returns comparable to the Contained regime during stagflationary periods, while FX trend underperforms (albeit still generating a positive average real return in Stagflation). The positive trend return in the Contained state is consistent with the results in Neville et al. (2021).

Exhibit 6 also provides results for a number of long-short equity strategies. An equally-weighted composite of seven different factor strategies provides average Stagflation returns that are higher than the baseline Contained state. Disaggregating, the Quality, Profitability, Investment, Size, and Bet-Against-Beta factors outperform in Stagflation.

5. Acceleration-Conditioned Asset Performance

5.1 The acceleration signal

A high inflation rate can persist while inflationary pressure is either intensifying or easing. These paths can have very different implications for expected monetary policy and asset prices. For an investor observing high inflation, whether the problem is getting worse or beginning to recede may therefore be as consequential as the regime classification itself. We examine this distinction using changes in realized inflation, which provide an observable measure of its direction but do not directly measure changes in market expectations.

Inflation is deemed to be accelerating when the CPI inflation rate over the past six months is above the CPI inflation rate over the six-month period prior to that, and decelerating otherwise. Similar to the approach for classifying high-inflation months, we apply the same two-month lag in order to capture publication delays in monthly CPI data for real-time utilization when classifying accelerating inflation months. The inflation acceleration signal is descriptive only; it plays no role in the regime definitions of **Sections 3-4**, and we compute it purely to split inflationary regimes into accelerating and decelerating sub-samples.

5.2 Results

We focus on Stagflation and Overheating to assess whether inflation acceleration adds useful context.

Exhibit 7. Acceleration-conditioned annualized real returns

Panel A. Equities, Bonds and the 60/40 Portfolio

Asset	Start Date	Stagflation		Overheating	
		Accelerating	Decelerating	Accelerating	Decelerating
Equities					
S&P 500	Jan-1919	-1.3%	2.5%	-8.0%	10.4%
S&P 500 (from Jul 1926)	*Jul-1926*	*0.6%*	*1.8%*	*-7.8%*	*11.1%*
Consumer Non-Durables	Jul-1926	7.8%	3.2%	-10.5%	13.1%
Consumer Durables	Jul-1926	-1.2%	4.4%	-5.4%	8.6%
Manufacturing	Jul-1926	-2.2%	3.6%	-8.8%	16.5%
Energy Equities	Jul-1926	3.8%	7.3%	6.5%	13.8%
Chemicals	Jul-1926	4.4%	1.9%	-11.0%	12.7%
Business Equipment	Jul-1926	1.1%	2.0%	-10.0%	18.6%
Telecom	Jul-1926	1.6%	-6.8%	-9.3%	6.8%
Utilities	Jul-1926	-2.8%	1.1%	-2.8%	5.8%
Wholesale and Retail	Jul-1926	12.4%	7.1%	-8.0%	10.0%
Healthcare	Jul-1926	7.2%	4.4%	-12.1%	11.5%
Finance	Jul-1926	7.7%	0.0%	-6.2%	12.6%
Other Industries	Jul-1926	0.2%	-1.9%	-13.1%	16.0%
Bonds					
US T-bills	Jan-1919	-0.6%	-0.5%	-3.3%	-0.3%
US 2-Year Treasury	Jan-1941	3.1%	0.4%	-6.5%	-0.8%
US 5-Year Treasury	Jan-1941	3.0%	-1.0%	-8.1%	0.1%
US 10-Year Treasury	Jan-1919	-1.0%	-2.1%	-7.3%	3.1%
US 30-Year Treasury	Jun-1953	1.3%	-4.1%	-13.7%	4.2%
US 10-Year TIPS	Apr-1959	-2.0%	5.5%	0.8%	4.6%
US Aaa Corporate	Jan-1919	-2.3%	-1.3%	-5.2%	3.6%
US Baa Corporate	Feb-1919	-3.5%	0.8%	-5.2%	3.5%
Diversified					
60/40 (S&P 500 / 10-Year Treasury)	Jan-1919	-1.2%	0.6%	-7.7%	7.5%

Panel B. Alternative assets

Asset	Start Date	Stagflation		Overheating	
		Accelerating	Decelerating	Accelerating	Decelerating
Commodities					
Broad Commodities (EW)	Apr-1929	-6.8%	2.0%	3.9%	7.0%
Gold	Feb-1975	16.9%	2.9%	17.9%	9.4%
Silver	Jul-1963	0.8%	9.7%	32.2%	14.6%
Precious Metals	Jul-1963	-5.3%	-2.4%	28.3%	12.0%
Agriculture	Aug-1929	4.2%	-3.9%	-1.2%	9.9%
Energy Futures	Apr-1983	-59.3%	7.8%	39.8%	-11.9%
Softs	Apr-1929	-1.1%	5.9%	0.6%	6.0%
Livestock	Jan-1965	1.6%	9.8%	-9.8%	12.9%
Industrial	Aug-1959	-20.0%	3.1%	15.6%	-1.3%
Real Estate					
US REITs	Jan-1972	9.5%	-11.1%	-0.7%	17.0%
US House Price Index	Apr-1953	-1.1%	-0.6%	1.6%	-2.3%
Trend-Following					
Multi-Asset Trend	Mar-1928	12.5%	14.3%	10.3%	3.6%
FX Trend	Apr-1971	-2.2%	6.4%	6.1%	-3.7%
Equity Trend	Mar-1928	2.8%	7.3%	-1.8%	8.7%
Commodity Trend	Feb-1947	17.0%	11.3%	13.3%	-1.1%
Bond Trend	Apr-1962	6.5%	9.3%	13.2%	5.3%
Long-Short Equity Factors					
Multi-Factor Equities (EW)	Jul-1963	7.5%	6.7%	3.0%	1.7%
Equity Momentum (WML)	Jul-1963	6.8%	10.7%	4.9%	-8.6%
Equity Value (HML)	Jul-1963	7.1%	0.4%	9.5%	-1.3%
Equity Profitability (RMW)	Jul-1963	5.9%	6.6%	0.1%	10.8%
Equity Investment (CMA)	Jul-1963	6.8%	3.2%	6.0%	-0.3%
Equity Quality (QMJ)	Jul-1963	6.2%	7.0%	5.0%	3.7%
Equity Size (SMB)	Jul-1963	13.6%	2.5%	-5.1%	6.3%
Equity Bet-Against-Beta (BAB)	Jul-1963	6.0%	16.3%	0.6%	0.9%

Source: Man Group, Finaeon, Bloomberg, Kenneth R. French Data Library, Shiller Data Library, Goldman Sachs, AQR. Jan 1919 to May 2026.

The acceleration separation has a dramatic impact on the Overheating regime. Accelerating inflation in Overheating regimes leads to very poor equity and bond returns. The 60-40 portfolio earns -7.7% p.a. in real terms. Interestingly, decelerating inflation is good for equity and bonds in the Overheating state; the 60-40 portfolio delivers an average real return of 7.5% p.a. Our acceleration signal leads to an extraordinary 15.2 percentage point difference in returns for the balanced portfolio.

Acceleration does not play as important a role for the Stagflation regime; this makes some sense because Overheating has the benefit of high economic growth. Similar to Overheating, accelerating inflation is generally worse than decelerating inflation in Stagflation. The 60-40 portfolio has a -1.2% p.a. real return in acceleration and a slightly better 0.6% return in deceleration. Overall, inflation acceleration is most important for separating good and bad Overheating periods.

The analysis of TIPS is interesting. Our simulated series goes back to 1959 so that it covers the key stagflationary episodes. Within Stagflation, TIPS performed much worse when inflation is accelerating than when it is decelerating. Indeed, even in Overheating environments, there was a substantially lower TIPS return during periods where inflation was accelerating.

There is a similar narrative for commodities. The average returns during periods of accelerating inflation are considerably lower than periods where inflation is decelerating. However, one important exception is gold. While constrained by a shorter history, gold tends to do better in accelerating inflation environments within both Stagflation and Overheating, consistent with the idea that investors flock to gold as a perceived safe haven in the worst economic environments.

Acceleration makes little difference for the performance of the multi-asset trend strategy; trend appears to be a reliable historical diversifier, delivering a positive real return on average in either state. A similar result is obtained for the equal-weighted long-short equity factor strategy. There is no significant difference in performance across both Stagflation and Overheating for these strategies after conditioning on inflation acceleration.

6. Insights from independent Stagflation episodes

The preceding analysis summarises average returns across many months and assets. Those averages are useful, but they can obscure the fact that individual Stagflation episodes differed sharply in origin, policy response and market path. The three case studies below are selected ex post as the longest Stagflation episodes confirmed by our real-time framework, ranked by duration rather than by asset outcome. They provide a complementary event-level perspective on how the same high-inflation, weak-growth label can contain very different investment environments.

Economic history and a systematic classification serve different purposes. We show the full historical windows, including the pre-confirmation run-up, and distinguish their asset returns from those exhibited during each confirmed Stagflation episode. The Vietnam War case study spans April 1966-October 1971, with 41 months before the September 1969 Stagflation episode; the OPEC embargo episode starts in August 1973, eight months before the Stagflation state is formally confirmed in April 1974; the second oil shock episode starts in October 1978, 12 months before confirmation of Stagflation in October 1979. The preceding periods do not always comprise a single regime: the Vietnam War example, in particular, changes state more than once. Regime classification uses a two-month macroeconomic data lag and three-month confirmation filter, so

these historical boundaries do not represent timely trade-entry signals, but instead offer a smooth, real-time signal that provides useful context for informing portfolio decisions.

6.1 Vietnam War and Great Society inflation, 1966-71

The April 1966-October 1971 historical window covers a largely demand-led inflation build up associated with spending relating to the Vietnam War and the Great Society initiative, followed by Federal Reserve tightening prior to the December 1969 to November 1970 recession. Before the 26-month Stagflation episode that ran from September 1969 to October 1971, the real-time signal oscillated between shorter Overheating and Stagflation states as industrial production growth fluctuated about its trailing trend level before persistently remaining below the trend level during the final Stagflation period. Extending the chart back to April 1966 makes both the build-up and the subsequent change in asset performance visible.

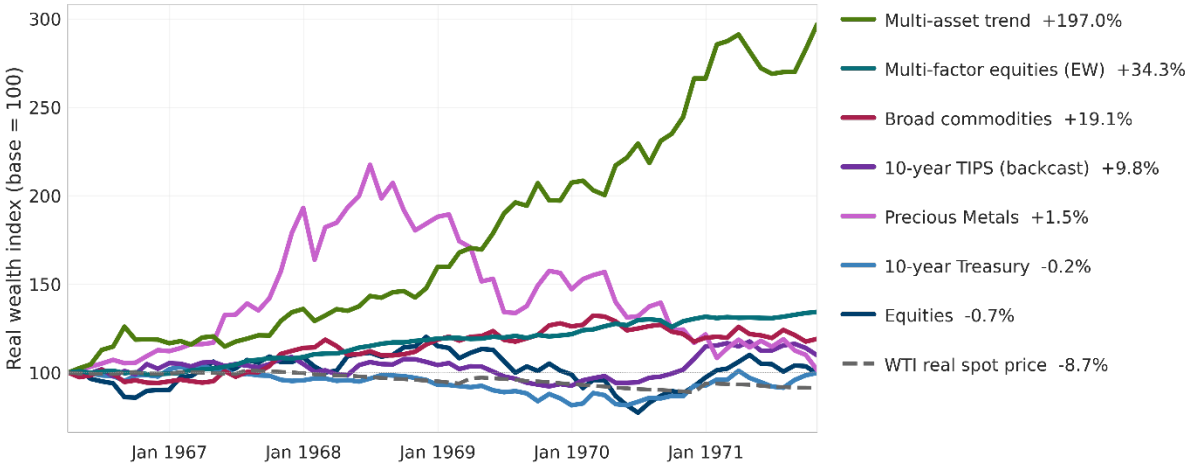
Returns differ substantially between the 41-month pre-confirmation window and the 26-month long confirmed run. During the run-up, Precious Metals gained 37.6%, broad commodities 19.3%, multi-factor equities 19.6% and multi-asset trend 94.3% in cumulative real terms. Additionally, 10-year Treasuries lost 11.8%, while equities gained 3.5% and the synthetic 10-year TIPS lost 5.7%.

During the confirmed September 1969-October 1971 run, Precious Metals gave back some of their initial gains, losing 26.3% real, broad commodities were near flat (-0.2%) and equities lost 4.0%. Nominal 10-year Treasuries instead gained 13.2% during the confirmed stagflationary episode, consistent with recession/disinflation forces offsetting inflation risk, although this descriptive sequence cannot be used to derive any definitive conclusions. TIPS gained 16.5%, multi-factor equities 12.3% and trend 52.8%.

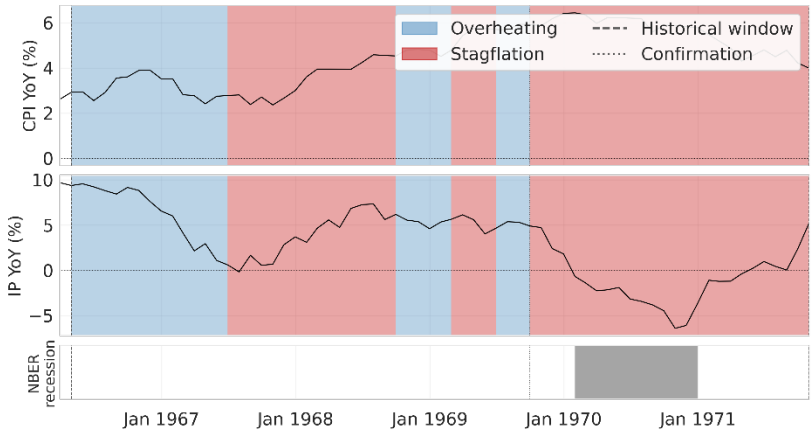
Compounding both periods, April 1966-October 1971 delivered 19.1% for broad commodities, 34.3% for multi-factor equities, 197.0% for trend and 9.8% for 10-year TIPS. Equities (-0.7%) and nominal Treasuries (-0.2%) ended near flat. Precious Metals finished up 1.5% despite their earlier surge. The synthetic TIPS series was not tradable during this episode, gold futures are unavailable, and the pre-1972 WTI spot series is shown for price context only because its long administered-price runs make a return comparison misleading.

Exhibit 8. Vietnam War / Great Society case study charts

Panel A. Cumulative real investable performance



Panel B. Five-year macro and regime context



Source: Man Group, Finaeon, Bloomberg, Kenneth R. French Data Library, AQR and NBER. Panel A shows cumulative real total returns over the confirmed September 1969-October 1971 Stagflation run, rebased to 100 at its start. Panel B shows how CPI and Industrial production evolved over a five-year window containing the stagflation episode; dotted vertical lines mark the confirmed run. Gold futures are unavailable.

6.2 The OPEC embargo, 1973-76

The August 1973-May 1976 window begins with eight months of classified Overheating before Stagflation is confirmed in April 1974, five months after the November 1973 recession began. The oil embargo and rollback of price controls added a pronounced supply dimension to already-elevated inflation (Blinder and Rudd, 2008). This sequence distinguishes an inflation shock while

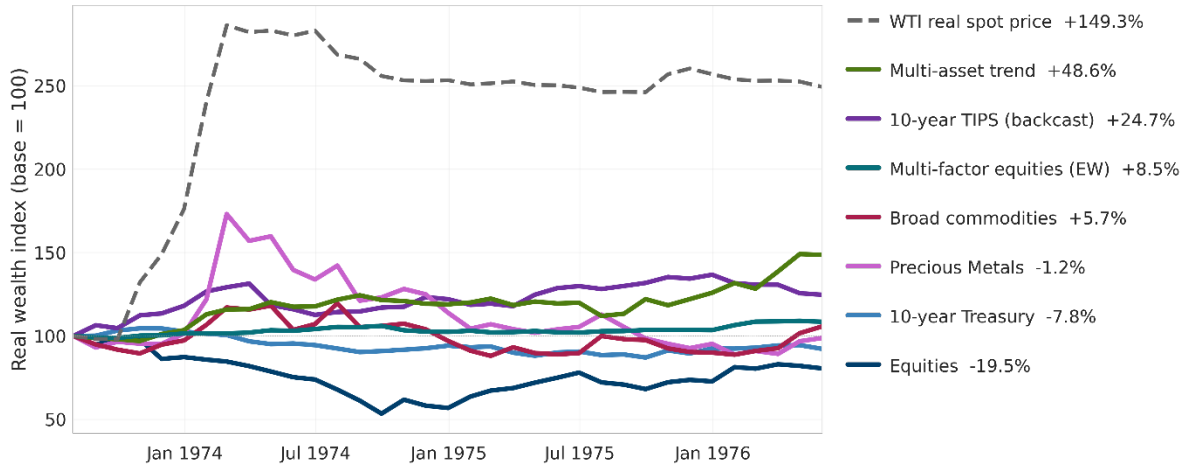
growth was still relatively resilient from the subsequent phase in which high inflation coincided with weaker production; it is not evidence that regime confirmation could have predicted the turning point. The confirmed Stagflation episode ends in May 1976.

In the eight-month run-up, equities and 10-year Treasuries lost 18.1% and 3.2% in real terms, respectively. Precious metals gained 57.0%, broad commodities 15.8%, 10-year TIPS 31.4% and multi-asset trend 16.0%. WTI's real spot price rose 182.1% over the same period, as oil prices surged in response to the supply shock. Once Stagflation was confirmed, precious metals lost 37.1% and broad commodities lost 8.7%, while multi-asset trend gained 28.1%. Equities and 10-year Treasuries lost a further 1.7% and 4.7% respectively, while 10-year TIPS lost 5.1% and multi-factor equities gained 6.3%.

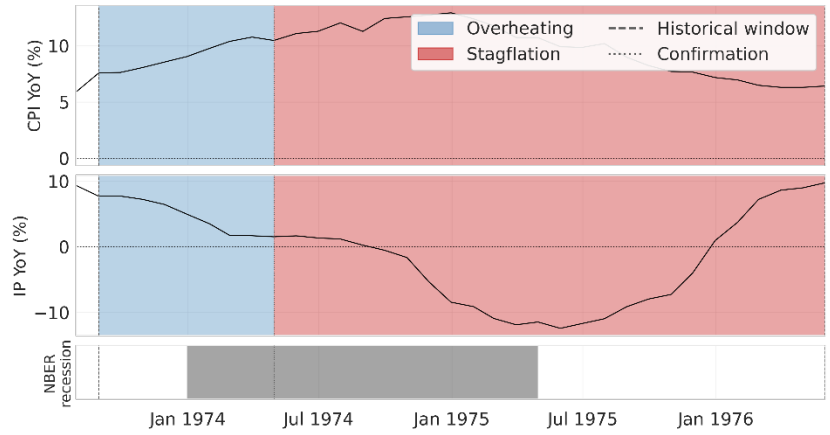
Over the full August 1973-May 1976 historical window, equities and Treasuries lost 19.5% and 7.8% respectively, while precious metals finished marginally negative in real terms (-1.2%), broad commodities gained 5.7%, TIPS gained 24.7% and trend gained 48.6%. The pre-1983 broad basket excludes Energy futures: its contrast with oil spot does not measure a tradable oil allocation. Gold futures begin in February 1975, overlapping with only 16 of the full window's 34-month sample, and are charted from that date but excluded from full-window comparisons. The WTI spot index rose 149.3% in real terms across the full window despite falling 11.6% during the confirmed Stagflation run.

Exhibit 9. OPEC embargo case study charts

Panel A. Cumulative real investable performance and WTI spot price context



Panel B. Five-year macro and regime context



Source: Man Group, Finaeon, Bloomberg, Kenneth R. French Data Library, AQR and NBER. Solid lines in Panel A are cumulative real investable total returns over the confirmed April 1974-May 1976 Stagflation run, rebased to 100 at its start. The dashed WTI line is a non-investable real spot-price index: it excludes futures roll, collateral yield, contract selection and implementation costs. The pre-1983 broad-commodity basket excludes Energy futures. Panel B shows how CPI and industrial production evolved over a five-year window containing the Stagflation episode; dotted vertical lines mark the confirmed run.

6.3 The second oil shock and Volcker disinflation, 1978-81

The October 1978-June 1981 window spans the Iranian-revolution oil shock, Volcker's appointment and the January-July 1980 recession, but ends before the recession beginning in July 1981. The classification records twelve months of Overheating (October 1978-September 1979) before confirming Stagflation for October 1979-June 1981. Policy tightening and inflation expectations could influence bond prices well before or after the confirmation date (Blinder and Rudd, 2008; Goodfriend and King, 2005). Comparing the two periods helps distinguish exposure to the initial price shock from exposure to later disinflation and weak activity, without attributing either return path solely to the regime label.

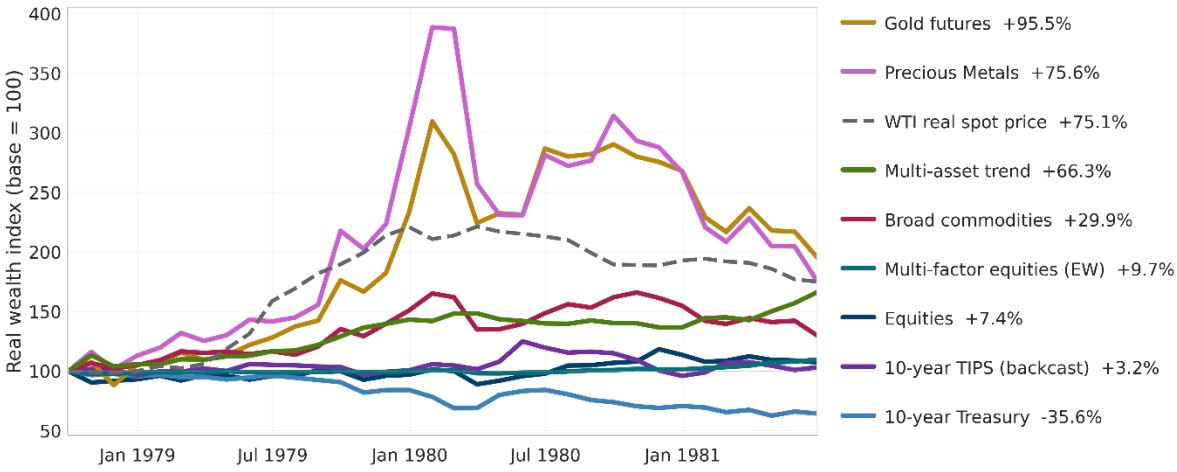
The Treasury sell-off was already under way during the 12-month Overheating run-up: 10-year nominal Treasuries lost 9.3% in real terms, while 10-year TIPS gained 3.5%. Gold futures gained 76.3%, precious metals 117.8%, broad commodities 35.3% and multi-asset trend 29.2%. Equities were broadly flat (-0.1%). During the confirmed 21-month Stagflation run, the 10-year Treasury lost another 29.0% even as equities gained 7.5%. 10-year TIPS fell by 0.3% real, precious metals lost 19.4% and broad commodities lost 4.0%. Gold futures still gained 10.9%, multi-factor equities 9.9% and trend 28.8%. Gold's divergence from the broader precious metals bucket reflects different constituent exposures, with the precious metals bucket containing more demand-sensitive, quasi-industrial commodities such as silver, platinum and palladium.

Over the full October 1978-June 1981 window, Treasury losses compound to 35.6%, against gains of 95.5% in gold, 75.6% in precious metals, 29.9% in broad commodities and 66.3% in trend. The real WTI spot index rose 75.1% over the full window but fell 7.7% during confirmed Stagflation;

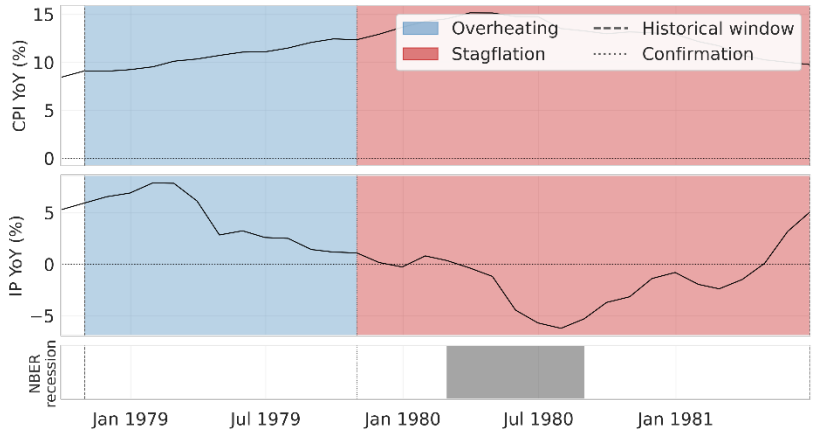
once again, those spot changes are price context, and do not represent investable returns. The switch from strong commodity gains in the run-up to weaker confirmed Stagflation run outcomes underscores the effect of timing, sector composition and policy repricing rather than a single 'oil-shock' return pattern.

Exhibit 10. Second oil shock / Volcker case study

Panel A. Cumulative real investable performance and WTI spot-price context



Panel B. Five-year macro and regime context



Source: Man Group, Finaeon, Bloomberg, Kenneth R. French Data Library, AQR and NBER. Solid lines in Panel A are cumulative real investable total returns over the confirmed October 1979-June 1981 Stagflation run, rebased to 100 at its start. The dashed WTI line is a non-investable real spot price index. The pre-1983 broad-commodity basket excludes Energy futures. Panel B shows how CPI and Industrial production evolved over a five-year window containing the stagflation episode; dotted vertical lines mark the confirmed run.

The longer historical windows show a recurrent distinction between inflationary build-up and the later long confirmed Stagflation run, but Vietnam War case study's 41-month run-up contains

several different confirmed states, not one continuous Overheating episode. Precious metals rose before Stagflation was confirmed and fell during it in all three cases, while broad commodities also lost ground in each confirmed Stagflation run. Nominal Treasuries differed sharply even within Stagflation: 13.2% in 1969-71 versus -29.0% in 1979-81. Across all 15 confirmed Stagflation episodes, equities and 10-year Treasuries were each positive in only seven; median cumulative real outcomes were -0.9% and -0.3%, with ranges -45.6% to 22.1% and -29.0% to 13.3%, respectively. These are descriptive, not independent experiments or formal significance tests. Trend and multi-factor equities gained across the selected historical windows, however this once again does not prove that these two asset classes will always be reliable Stagflation hedges.

These histories illustrate how much the investment experience depends on the path into and through a regime. Commodity gains were concentrated in the run-up to the selected Stagflation episodes, while equity returns within the confirmed periods could combine deep drawdowns with subsequent recoveries. An allocation made after confirmation would therefore have faced a different opportunity set from one established during the initial inflationary shock. The regime label alone is not sufficient to resolve that timing decision.

7. Conclusions and implications for asset managers

We already know from Neville et al. (2021) that inflationary states are bad for stocks and bonds. However, we argue that it is useful to separate the high inflation episodes into two different regimes: Stagflation and Overheating. There are immediate insights. For example, our results show that fixed income performs particularly poorly in the Overheating state. Our results also show variation in equity performance across sectors that could be useful for asset managers defensively repositioning their portfolio conditional on our real-time signal.

Conditioning on inflation's direction produces a particularly large separation in US Overheating returns. Using our six-month measure, the 60/40 portfolio's average annualized real return is −7.7% when inflation accelerates and 7.5% when it decelerates. This suggests that investors should consider whether inflationary pressure is intensifying or easing alongside the prevailing growth-inflation regime. The result is sensitive to the measurement horizon, however, and the corresponding separation is smaller in US Stagflation. These conditional averages do not establish a profitable switching strategy.

We also analyze the diversifiers. Our results point to gold, trend, and long-short equity factor strategies as potentially providing material diversification. Notably, the diversification ability is robust across both Stagflation and Overheating as well as the acceleration/deceleration subsamples. Importantly, we provide analysis of individual strategies and there is variation in performance for the factor strategies with quality-like strategies providing the most secure historical protection. We also highlight the role of gold in a diversified portfolio. However, in contrast to other assets, gold has a limited history given that it was not legal for investors to hold between 1933 and 1974. As a result, some important episodes are omitted. TIPS appear to have been a reliable hedge for both Stagflation and Overheating episodes.

Finally, we want to emphasize a key aspect of our research strategy. The standard approach is to look at asset returns in ex-post Stagflation and Overheating states. This type of analysis assumes knowledge of the future. For example, it assumes the investor knows the NBER peaks and troughs before the NBER makes the announcements. This is not what we do. We present real-time indicators of both growth and inflation. While our data do reflect revisions in the data, we argue that these revisions are minor. Given that our regime designation is in real time, there may be some incorrect (ex-post) classifications. For asset managers, the framework provides context for assessing portfolio exposures. Using it to change allocations requires a further judgement about what markets have already priced in and whether the direction of inflation justifies acting before the economic regime itself changes.

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Appendix A: International Evidence — United Kingdom and Japan

A1. Design

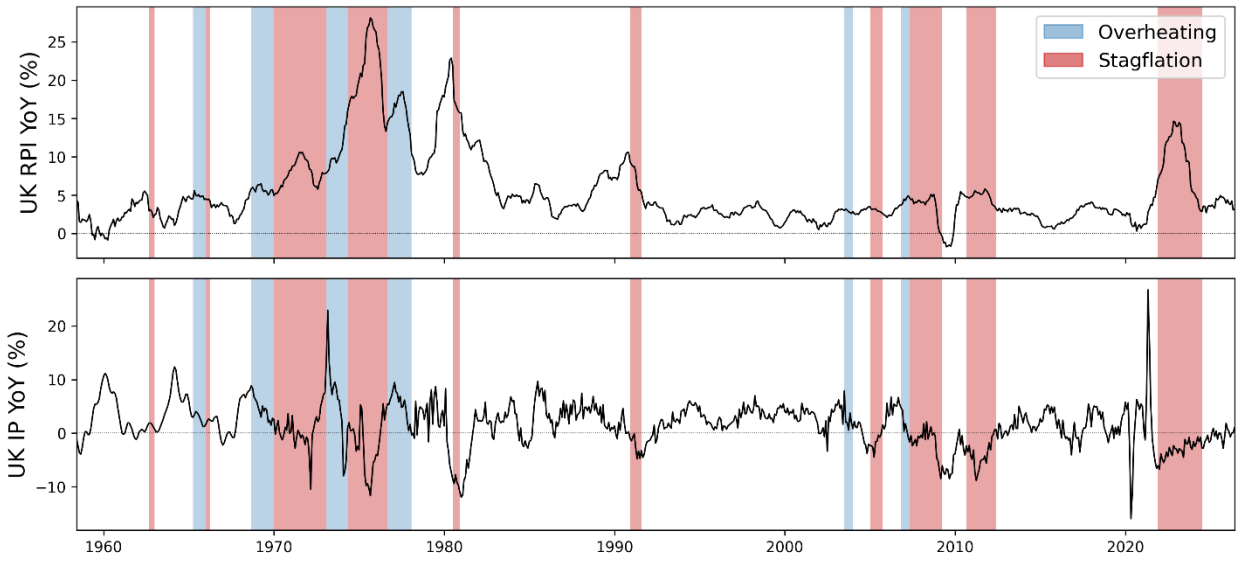
We apply the same three-regime classification framework to the United Kingdom (UK) and Japan to examine whether similar cross-regime patterns arise outside the US. The inflation signal uses local seasonally adjusted RPI for the UK and CPI for Japan (both sourced from Finaeon), while the growth signal uses local industrial production indices sourced from the Office for National Statistics (ONS) and Japan's Ministry of Economy, Trade and Industry (METI).

As in the US study, country-specific CPI/RPI and industrial production signals are delayed by two months, and a state change is recognized only on its third consecutive candidate month. The two-month delay is harmonized for comparability; it is not a claim that UK and Japanese publication calendars are identical. Historical macroeconomic series are revised, so the design is pseudo-real-time rather than vintage-data real time.

The asset universe is intentionally narrower than in the US study: local equities, local 10-year government bond returns and a 60/40 stock-bond portfolio, all deflated by the local seasonally adjusted inflation series. Bond returns use the same yield-based approximation as in the US analysis. After the necessary warm-up, the usable classification samples begin in May 1958 for the UK and May 1967 for Japan.

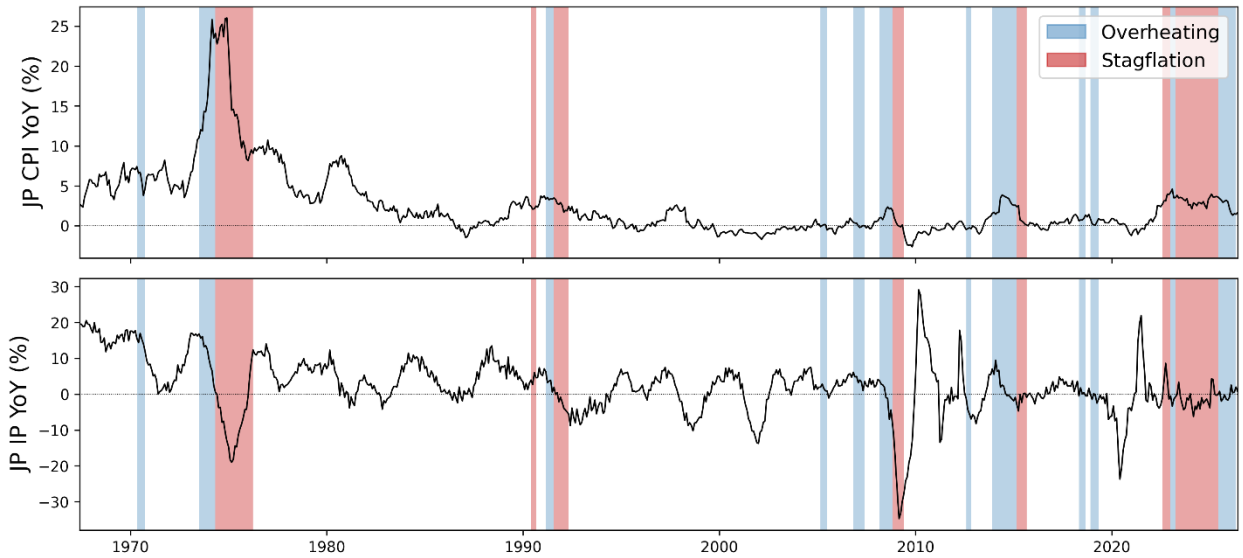
A2. Regime Incidence

Exhibit A1. UK: YoY RPI and IP with Overheating and Stagflation regimes shaded



Source: Man Group, Finaeon, ONS. Blue = Overheating, Red = Stagflation. The chart displays raw macroeconomic data from January 1958; regime classification and reported results use the May 1958-May 2026 usable sample.

Exhibit A2. Japan: YoY CPI and IP with Overheating and Stagflation regimes shaded



Source: Man Group, Finaeon, METI. Blue = Overheating, Red = Stagflation. The chart displays raw macroeconomic data from January 1967; regime classification and reported results use the May 1967-May 2026 usable sample.

Table A1. Regime distribution

Region	Stagflation	Overheating	Contained	Total
UK	169	69	579	817
Japan	79	80	550	709

Source: Man Group. UK data runs from May 1958 to May 2026 and Japan data runs from May 1967 to May 2026.

A3. Regime-Conditional Performance

Table A2. Annualised real returns by regime

Region	Asset	Start Date	Average Real Returns (% p.a.)		
			Stagflation	Overheating	Contained
UK	Equity	May-1958	5.8%	-5.0%	9.6%
UK	10yr Bond	May-1958	1.1%	-5.0%	3.6%
UK	60/40	May-1958	3.9%	-5.0%	7.2%
Japan	Equity	May-1967	0.3%	0.3%	8.6%
Japan	10yr Bond	May-1967	-1.2%	-3.4%	3.3%
Japan	60/40	May-1967	-0.3%	-1.1%	6.5%

Source: Man Group, Finaeon, Bloomberg.

The international results are heterogeneous rather than a simple replication of the US pattern. UK assets have positive average real returns during Stagflation but weaker outcomes during

Overheating than in Contained months. In Japan, we identify relative weakness in both Stagflation and Overheating environments, with the Overheating environment once again being especially punitive for both bonds and equities. These reduced, country-specific panels are best read as an external descriptive check across differing institutional and macroeconomic settings, not as a universal cross-country ranking.

A4. Acceleration-conditioned asset performance

Table A3. Acceleration-conditioned annualised real returns

Region	Asset	Start Date	Stagflation		Overheating	
			Accelerating	Decelerating	Accelerating	Decelerating
UK	Equity	May-1958	-2.3%	12.4%	-12.3%	3.9%
UK	10yr Bond	May-1958	-5.0%	6.1%	-10.2%	1.3%
UK	60/40	May-1958	-3.4%	9.9%	-11.5%	2.9%
Japan	Equity	May-1967	-4.2%	3.9%	-3.6%	5.0%
Japan	10yr Bond	May-1967	-1.1%	-1.3%	-6.3%	0.1%
Japan	60/40	May-1967	-3.0%	1.8%	-4.7%	3.0%

Source: Man Group, Finaeon, Bloomberg.

Appendix B: Data and Sensitivity Checks for the US Study

This appendix outlines the approach used to seasonally adjust the inflation series, reports sensitivity checks for the high-inflation threshold, the trailing reference window and the inflation-acceleration horizon and also includes an asset data summary table. The sensitivity exercises assess the sensitivity of our classification signals to changes in the underlying parameters rather than a search for the parameterization with the strongest return result.

B1. Seasonal adjustment of US CPI

Seasonal adjustment is itself point-in-time. Let p_t denote the monthly non-seasonally adjusted CPI level at time t and let $\pi_t = log(p_t) - log(p_{t-1})$ denote the (log-based) monthly inflation rate. We apply a walk-forward seasonal adjustment (that uses a rolling 10-year window to derive seasonal adjustment factors) to the non-seasonally adjusted series as follows:

At each month-end, t (after applying a 120-month warmup period), we define a trailing window, $W_t = \{t - 120, \dots, t - 1\}$.

The seasonal factor for month t's calendar month, $m(t)$, is:

$$s_t = mean\{\pi_\tau : \tau \in W_t, m(\tau) = m(t)\} - mean\{\pi_\tau : \tau \in W_t\}$$

Where π_t is the log-based monthly inflation rate at time t.

We then calculate seasonally adjusted inflation rates, π_t^{SA}, as $\pi_t^{SA} = \pi_t^{NSA} - s_t$ and re-construct our seasonally adjusted CPI index, CPI_t^{SA}, as $CPI_t^{SA} = CPI_{t-1}^{SA} \exp(\pi_t^{SA})$.

Using a walk-forward approach based on rolling 10-year windows helps avoid lookahead bias (which would happen if we instead used full sample averages) and ensures that the seasonal adjustment factors change through time and therefore capture any changes in seasonal patterns.

B2. Asset data

Table B1. Asset data summary table

The following asset data summary table reports the usable start date and regime coverage for each series used in the US analysis. Sample lengths differ because return histories begin at different dates.

Asset	Start Date	Sample count			
		Stagflation	Overheating	Contained	Total
S&P 500	Jan-1919	172	186	931	1289
S&P 500 (from Jul 1926)	Jul-1926	156	179	864	1199
Consumer Non-Durables	Jul-1926	156	179	864	1199

Consumer Durables	Jul-1926	156	179	864	1199
Manufacturing	Jul-1926	156	179	864	1199
Energy Equities	Jul-1926	156	179	864	1199
Chemicals	Jul-1926	156	179	864	1199
Business Equipment	Jul-1926	156	179	864	1199
Telecoms	Jul-1926	156	179	864	1199
Utilities	Jul-1926	156	179	864	1199
Wholesale and Retail	Jul-1926	156	179	864	1199
Healthcare	Jul-1926	156	179	864	1199
Finance	Jul-1926	156	179	864	1199
Other Industries	Jul-1926	156	179	864	1199
US T-bills	Jan-1919	172	186	931	1289
US 2-Year Treasury	Jan-1941	149	141	735	1025
US 5-Year Treasury	Jan-1941	149	141	735	1025
US 10-Year Treasury	Jan-1919	172	186	931	1289
US 30-Year Treasury	Jun-1953	145	98	633	876
US 10-Year TIPS	Apr-1959	145	98	563	806
US Aaa Corporate	Jan-1919	172	186	931	1289
US Baa Corporate	Feb-1919	171	186	931	1288
Gold	Feb-1975	90	68	458	616
Silver	Jul-1963	145	98	512	755
Broad Commodities (EW)	Apr-1929	156	179	831	1166
Precious Metals	Jul-1963	145	98	512	755
Agriculture	Aug-1929	156	179	827	1162
Energy	Apr-1983	53	56	409	518
Softs	Apr-1929	156	179	831	1166
Livestock	Jan-1965	145	98	494	737
Industrial	Aug-1959	145	98	559	802
Multi-Asset Trend	Mar-1928	156	179	844	1179
FX Trend	Apr-1971	107	76	479	662
Equity Trend	Mar-1928	156	179	844	1179
Commodity Trend	Feb-1947	145	114	693	952
Bond Trend	Apr-1962	145	98	527	770
Multi-Factor Equities (EW)	Jul-1963	145	98	512	755
Equity Momentum (WML)	Jul-1963	145	98	512	755

Equity Value (HML)	Jul-1963	145	98	512	755
Equity Profitability (RMW)	Jul-1963	145	98	512	755
Equity Investment (CMA)	Jul-1963	145	98	512	755
Equity Quality (QMJ)	Jul-1963	145	98	512	755
Equity Size (SMB)	Jul-1963	145	98	512	755
Equity Bet-Against-Beta (BAB)	Jul-1963	145	98	512	755
60/40 (S&P / 10yr)	Jan-1919	172	186	931	1289

Source: Man Group, Finaeon, Bloomberg, Kenneth R. French Data Library, Goldman Sachs, AQR.

B3. Inflation Threshold Sensitivity

The base definition classifies high inflation when CPI year-on-year inflation exceeds its trailing 10-year 75th percentile. Holding the growth rule, two-month information delay and three-month regime confirmation filter fixed, the table varies only that percentile.

Table B2. Regime classification results for different CPI percentile thresholds

CPI percentile	Stagflation Months	Average CPI YoY (%p.a.)	Average IP YoY (%p.a.)	S&P 500 real returns (%p.a.)	10yr Treasuries real returns (%p.a.)
60th	253	6.4%	-1.3%	1.1%	-0.6%
65th	228	6.8%	-1.3%	1.5%	-0.2%
70th	198	6.9%	-1.4%	2.9%	-0.9%
75th (base)	172	7.2%	-1.6%	0.7%	-1.6%
80th	136	7.5%	-2.4%	-1.2%	-1.5%
85th	112	8.0%	-3.1%	-3.4%	-2.0%

Source: Man Group, Finaeon. Macro data are lagged by two months and three-month regime confirmation filter is applied to regime classification framework. Jan 1919 to May 2026.

The table above shows that tightening the threshold reduces the sample and raises average inflation (by construction). 10-year Treasury returns are negative at every threshold. Equity is positive at all but the 80th and 85th percentile thresholds.

B4. Trailing Window Horizon Sensitivity

The base specification uses a 10-year window for both the CPI percentile and IP-CAGR reference values. The sensitivity varies both windows while retaining the two-month macro delay and three-month confirmation rule.

Table B3. Regime classification results for different trailing window horizons

Window	Stagflation months	Average YoY CPI (%p.a.)	Average YoY IP (%p.a.)	S&P 500 real returns (%p.a.)	10yr Treasuries real returns (%p.a.)
5 years	182	5.7%	-2.1%	0.7%	1.9%
10 years (base)	172	7.2%	-1.6%	0.7%	-1.6%
20 years	128	7.6%	-0.7%	-3.4%	-3.4%

Source: Man Group, Finaeon. Macro data are lagged by two months and three-month regime confirmation filter is applied to regime classification framework . Jan 1919 to May 2026.

The 20-year specification has 1,265 usable months rather than the 1,289 from the main study because of the longer warm-up period.

B5. Inflation Acceleration Horizon

The headline acceleration condition compares adjacent, non-overlapping six-month cumulative CPI changes. The sensitivity results below repeat the acceleration-versus-deceleration split separately within the Stagflation and Overheating regimes. The asset list is deliberately limited to a subsample of the core assets used in the main discussion.

Table B4. Acceleration-conditioned annualised real returns across different horizons

Panel A. 3-month horizon

Asset	Stagflation		Overheating	
	Accelerating	Decelerating	Accelerating	Decelerating
S&P 500	1.8%	-0.3%	3.3%	-3.5%
US 10-Year Treasury	4.9%	-7.5%	-3.2%	-2.7%
Gold	-9.5%	21.4%	18.7%	9.1%
Broad Commodities (EW)	-10.7%	5.7%	5.6%	4.8%
Multi-Asset Trend	13.2%	13.7%	8.5%	6.4%
Multi-Factor Equities (EW)	7.2%	6.9%	-0.5%	5.5%
60/40 (S&P / 10yr)	3.0%	-3.2%	0.7%	-3.2%

Panel B. 6-month horizon

Asset	Stagflation		Overheating	
	Accelerating	Decelerating	Accelerating	Decelerating
S&P 500	-1.3%	2.5%	-8.0%	10.4%
US 10-Year Treasury	-1.0%	-2.1%	-7.3%	3.1%
Gold	16.9%	2.9%	17.9%	9.4%
Broad Commodities (EW)	-6.8%	2.0%	3.9%	7.0%

Multi-Asset Trend	12.5%	14.3%	10.3%	3.6%
Multi-Factor Equities (EW)	7.5%	6.7%	3.0%	1.7%
60/40 (S&P / 10yr)	-1.2%	0.6%	-7.7%	7.5%

Panel C. 9-month horizon

Asset	**Stagflation**		**Overheating**	
	Accelerating	**Decelerating**	**Accelerating**	**Decelerating**
S&P 500	-3.3%	4.3%	-6.0%	12.7%
US 10-Year Treasury	-3.9%	0.5%	-4.2%	0.0%
Gold	23.1%	-0.2%	12.9%	17.1%
Broad Commodities (EW)	-6.2%	1.6%	4.7%	6.3%
Multi-Asset Trend	12.4%	14.4%	10.0%	1.8%
Multi-Factor Equities (EW)	5.2%	9.0%	4.5%	-4.1%
60/40 (S&P / 10yr)	-3.5%	2.8%	-5.3%	7.6%

Source: Man Group, Finaeon, Bloomberg.